

• Drones are helping farmers scan fields, monitor crops, seeding and analyse plant health, among others. Drones with advanced sensors and image-processing capabilities can capture highly accurate images of the fields, and expose everything from irrigation problems and soil variation to pest and fungal infestations by seeing crops from the air.

Drones are also being used to spray micronutrients on crops while being careful not to penetrate groundwater. Recent studies have shown that drones can increase the speed of spraying by five times compared to other types of machinery.

India has recently unveiled its Drone Regulation policy. These guidelines will foster technology and innovation in the development of drones that have an extensive range of applications ranging from disaster relief to agriculture. A Kerala-based company and a leading drone manufacturing company in south India are working together with Kerala government on a precision agriculture project to monitor farms and crop health using drones.

Data-enabled agriculture combines application of sensor connectivity, drone, drone storage and aggregation, optimisation hardware, software platforms, Big Data analytics and the IoT, converting traditional farming into smart farming.

The Indian government is serious about leveraging new-era digital technologies for agriculture. In May 2019, NITI Aayog, the government's main think-tank, signed an agreement with one of the big software firms to develop a model for crop yield predictions using AI so that farmers can be provided real-time advisories.

The number of people who will be using the Internet by 2020 is expected to be 700 million. This would be the inflection point for the ag-tech market in India. Agriculture startups can unleash umpteen opportunities to strengthen the supply chain in agriculture.

Common barriers to the commercialisation and scaling up of technology are access to finance, cash flow management, gaps in technology infrastructure, limited access to institutionalised farmer networks and others. For innovation and entrepreneurship to be effective in transforming agriculture in India, it is important to address these issues and create an enabling environment.

Summary

Uptake of technologies at market prices in a sector that has traditionally been heavily subsidised remains challenging, but farmers are prompt to identify what works in their interest and are ready to pay for it. Digital technologies offer the potential to achieve necessary conditions for scale, with distributed low-cost and customised delivery, thus attracting private investments in agri-innovation.

The challenge before India lies in balancing high growth with inclusive growth in agriculture, which can be achieved by an effective policy framework, innovative financing and enabling a technology ecosystem, which can empower Indian farmers and make agriculture sustainable. **EFY**

BINAY – Lighting Up The Future

BINAY

LED Aviation Obstruction

Lights

High Intensity
200,000 candela
(Flashing White LEDs)

Medium Intensity
20,000 candela
(Flashing White LEDs)

Medium Intensity
2,000 candela
(Flashing Red
OR Fixed Red LEDs)

BINAY LED-based HIGH, MEDIUM and LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime

All BINAY Aviation Lights are based on Non-switching passive type circuitry (NO SMPS)

The circuitry incorporated inside the LED Aviation Obstruction Lights is passive electronic (solid-state) in nature, and does not use internal SMPS driver switching units to control the light unit. Internal LED current control is by non-switching electronic means only. No electrolytic capacitors are used inside the Aviation Obstruction Light unit.

THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT

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